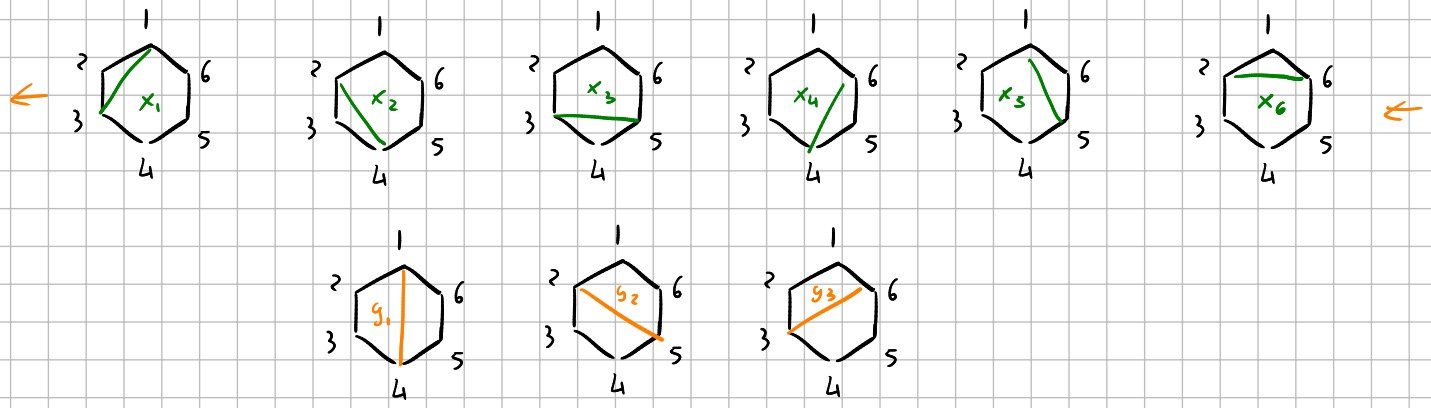
$n = 6$

$$\overset{x_1}{X_{13}} \to \overset{x_2}{X_{24}} \to \overset{x_3}{X_{35}} \to \overset{x_4}{X_{46}} \to \overset{x_5}{X_{51}} \to \overset{x_6}{X_{62}}$$

$$\overset{y_1}{X_{14}} \to \overset{y_2}{X_{25}} \to \overset{y_3}{X_{36}}$$



<u>interacting:</u> $x_1 \overset{x_6}{x_0}$ $\quad x_1^2 \quad x_1 x_2 \quad x_1 y_2$

$x_2 y_3$

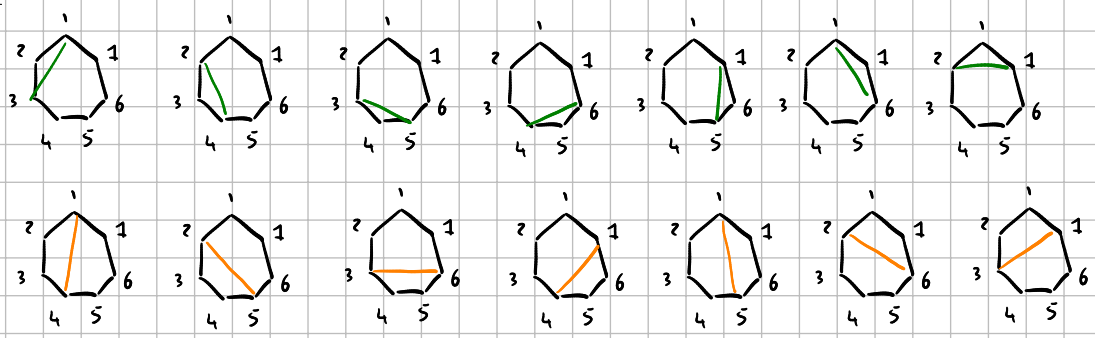
$x_3 y_1 \equiv x_3 y_4$

$x_4 y_2 \equiv x_4 y_5$

$$\boxed{\begin{array}{l} x_i^2 \quad x_{i\,i\pm1} \\ x_i\, y_{i+1} \\ y_i y_j \quad \forall i,j \end{array}}$$

non local terms

$n = 7$



$x_i^2 \quad x_{i\,i\pm1}$ non local

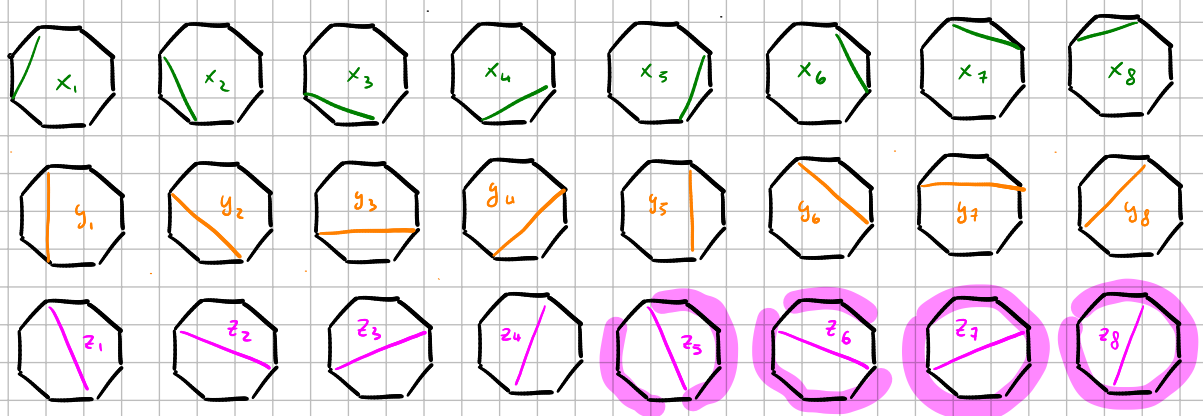
$x_1 y_2$ non local

$x_1 y_6$ non local

$y_1^2 \quad y_1 y_2 \quad y_1 y_3 \quad y_1 y_7 \quad y_1 y_6$

$$\boxed{\begin{array}{l} x_i^2 \quad x_{i\,i\pm1} \\ x_i\, y_{i+1} \\ x_i\, y_{i+5} \quad \text{or} \quad x_i\, y_{i-2} \\ \\ y_i^2 \quad y_i y_{i\pm1} \quad y_i y_{i\pm2} \end{array}}$$

non local



$\boxed{x_i^2 \quad x_i^2 x_{i\pm1}}$ $\quad x_1 y_2 \quad x_1 y_7 \quad$ so $\quad \boxed{x_i y_{i+1}} \quad \boxed{x_i y_{i-2}}$

$y_1 y_7 \quad y_1 y_8 \quad y_1^2 \quad y_1 y_2 \quad y_1 y_3$

$\boxed{y_i y_{i\pm1} \quad y_i^2 \quad y_i y_{i\pm2}}$

$x_1 z_2 \quad x_1 z_{1-3}$

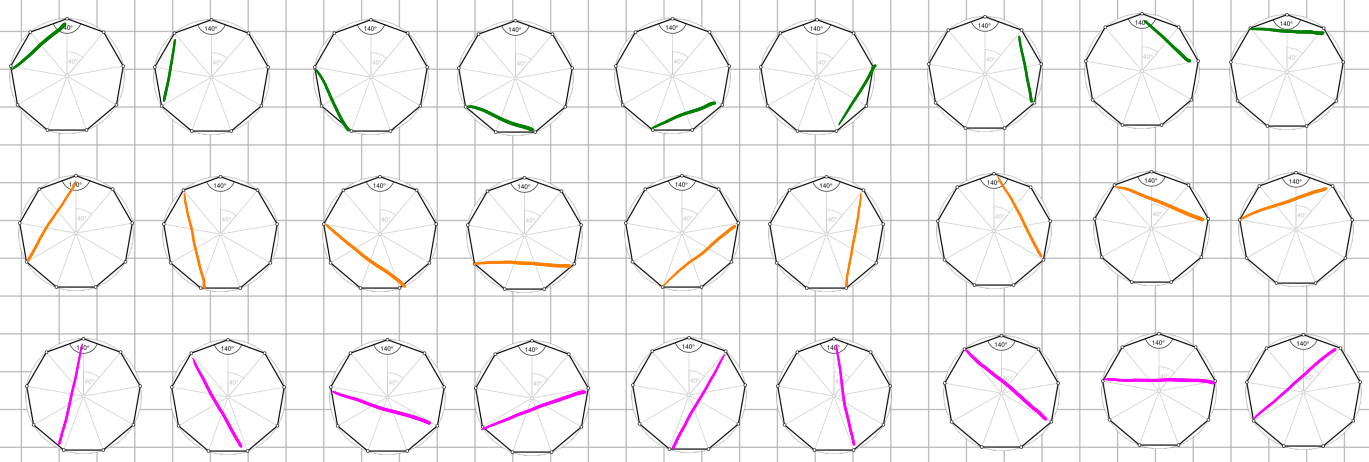
$\boxed{x_i z_{i+1} \quad x_i z_{i-3}}$

$z_1^2 \quad z_1 z_2 \quad z_1 z_3 \quad z_1 z_4$

$\boxed{z_i^2 \quad z_i z_{i\pm1} \quad z_i z_{i\pm2} \quad z_i z_{i\pm3}}$

$y_1 z_2 \quad y_1 z_3 \quad y_1 z_6 \quad y_1 z_7$

$\boxed{\begin{array}{ll} y_i z_{i+1} & y_i z_{i+2} \\ y_i z_{i-2} & y_i z_{i-3} \end{array}}$



$x_1 y_2 \quad x_1 y_8$

$x_1 z_2 \quad x_1 z_7$

$\boxed{\begin{array}{ll} x_i y_{i+1} & x_i y_{i-2} \\ x_i z_{i+1} & x_i z_{i-3} \end{array}}$

$y_1 y_{-1} \quad y_1 y_0 \quad y_1^2 \quad y_1 y_2 \quad y_1 y_3$

$z_1 z_{-2} \quad z_1 z_{-1} \quad z_1 z_0 \quad z_1^2 \quad z_1 z_2 \quad z_1 z_3 \quad z_1 z_4$

$y_1 z_2 \quad y_1 z_3 \quad y_1 z_7 \quad y_1 z_8$

$\boxed{\begin{array}{ll} y_i z_{i+1} & y_i z_{i+2} \\ y_i z_{i-2} & y_i z_{i-3} \end{array}}$

n	X	Y	Z	W	# groups
4	$\boxed{2}$	0	0		1
5	5	0	0		1
6	6	$\boxed{3}$	0		2
7	7	7	0		2
8	8	8	$\boxed{4}$		3
9	9	9	9		3
10	10	10	10	5	4
11	11	11	11	11	4

guess

$\boxed{\text{1-zero}}$

$n=4$

$$\begin{cases} x_{24} = -x_{13} \end{cases} \qquad \begin{cases} x_2 = -x_1 \end{cases}$$

$13 \to 24 \to 35 \to 41 \to 52$

$n=5$

$$\begin{cases} x_{24} = x_{14} - x_{13} \\ x_{25} = -x_{13} \\ x_{35} \text{ arbitrary} \end{cases} \qquad \begin{cases} x_2 = x_4 - x_1 \\ x_5 = -x_1 \\ x_3 \text{ arbitrary} \end{cases}$$

x	13	24	35	46	51	62
y	14	25	36			

$n=6$

$$\begin{cases} x_{24} = x_{14} - x_{13} \\ x_{25} = x_{15} - x_{13} \\ x_{26} = -x_{13} \\ x_{35}\ x_{36}\ x_{46} \text{ arbitrary} \end{cases} \qquad \begin{cases} x_2 = y_1 - x_1 \\ y_2 = x_5 - x_1 \\ x_6 = -x_1 \\ x_3\ x_4\ y_3 \text{ arbitrary} \end{cases}$$

13	24	35	46	57	61	72
14	25	36	47	51	62	73

$n=7$

$$\begin{cases} x_{24} = x_{14} - x_{13} \\ x_{25} = x_{15} - x_{13} \\ x_{26} = x_{16} - x_{13} \\ x_{27} = -x_{13} \\ x_{35}\ x_{36}\ x_{37}\ x_{46}\ x_{47}\ x_{57} \text{ arbitrary} \end{cases} \qquad \begin{cases} x_2 = y_1 - x_1 \\ y_2 = y_5 - x_1 \\ y_6 = x_6 - x_1 \\ x_7 = -x_1 \\ x_3\ x_4\ x_5\ y_3\ y_4\ y_7 \text{ arbitrary} \end{cases}$$

13	24	35	46	57	68	71	82
14	25	36	47	58	61	72	83
15	26	73	84				

$n=8$

$$\begin{cases} x_{24} = x_{14} - x_{13} \\ x_{25} = x_{15} - x_{13} \\ x_{26} = x_{16} - x_{13} \\ x_{27} = x_{17} - x_{13} \\ x_{28} = -x_{13} \\ x_{35}\ x_{36}\ x_{37}\ x_{38} \\ x_{46}\ x_{47}\ x_{48} \\ x_{57}\ x_{58} \\ x_{68} \text{ arbitrary} \end{cases} \qquad \begin{cases} x_2 = y_1 - x_1 \\ y_2 = z_1 - x_1 \\ z_2 = y_6 - x_1 \\ y_7 = x_7 - x_1 \\ x_8 = -x_1 \\ x_3\ x_4\ x_5\ x_6 \\ y_3\ y_4\ y_5 \text{ arbitrary} \\ z_3\ z_4 \\ y_8 \end{cases}$$

13	24	35	46	57	68	79	81	92
14	25	36	47	58	69	71	82	93
15	26	37	48	59	61	72	83	94
1	2	3	4	5	6	7	8	9

$n = 9$

$$\begin{cases} x_{24} = x_{14} - x_{13} \\ x_{25} = x_{15} - x_{13} \\ x_{26} = x_{16} - x_{13} \\ x_{27} = x_{17} - x_{13} \\ x_{28} = x_{18} - x_{13} \\ x_{29} = -x_{13} \end{cases}$$

$$\begin{array}{ccccc} x_{35} & x_{36} & x_{37} & x_{38} & x_{39} \\ & x_{46} & x_{47} & x_{48} & x_{49} \\ & & x_{57} & x_{58} & x_{59} \\ & & & x_{68} & x_{69} \\ & & & & x_{79} \end{array}$$

$$\begin{cases} x_2 = y_1 - x_1 \\ y_2 = z_1 - x_1 \\ z_2 = z_6 - x_1 \\ z_7 = y_7 - x_1 \\ y_8 = x_8 - x_1 \\ x_9 = -x_1 \end{cases}$$

$x_3 \; x_4 \; x_5 \; x_6 \; x_7$

$y_3 \; y_4 \; y_5 \; y_6$

$z_3 \; z_4 \; z_5$

$z_8 \; z_9$

y_9

$$\begin{cases} x_2 = y_1 - x_1 \\ y_2 = z_1 - x_1 \\ z_2 = y_6 - x_1 \\ y_7 = x_7 - x_1 \\ x_8 = -x_1 \end{cases}$$

$n = 8$

4

$$\{ x_2 = -x_1$$

5

$$\begin{cases} x_2 = x_4 - x_1 \\ x_5 = -x_1 \\ x_3 \quad \text{arbitrary} \end{cases}$$

6

$$\begin{cases} x_2 = y_1 - x_1 \\ y_2 = x_5 - x_1 \\ x_6 = -x_1 \\ x_3 \; x_4 \; y_3 \quad \text{arbitrary} \end{cases}$$

7

$$\begin{cases} x_2 = y_1 - x_1 \\ y_2 = y_5 - x_1 \\ y_6 = x_6 - x_1 \\ x_7 = -x_1 \\ x_3 \; x_4 \; x_5 \; y_3 \; y_4 \; y_7 \end{cases}$$

8

$$\begin{cases} x_2 = y_1 - x_1 \\ y_2 = z_1 - x_1 \\ z_2 = y_6 - x_1 \\ y_7 = x_7 - x_1 \\ x_8 = -x_1 \\ x_3 \; x_4 \; x_5 \; x_6 \\ y_3 \; y_4 \; y_5 \quad \text{arbitrary} \\ z_3 \; z_4 \\ y_8 \end{cases}$$

9

$$\begin{cases} x_2 = y_1 - x_1 \\ y_2 = z_1 - x_1 \\ z_2 = z_6 - x_1 \\ z_7 = y_7 - x_1 \\ y_8 = x_8 - x_1 \\ x_9 = -x_1 \\ x_3 \; x_4 \; x_5 \; x_6 \; x_7 \\ y_3 \; y_4 \; y_5 \; y_6 \\ z_3 \; z_4 \; z_5 \\ z_8 \; z_9 \\ y_9 \end{cases}$$

(4)

$$\begin{cases} \boxed{x_2 = -x_1} \end{cases}$$

(5)

$$\begin{cases} x_2 = x_4 - x_1 \\ \boxed{x_5 = -x_1} \\ x_3 \quad \text{arbitrary} \end{cases}$$

(6)

$$\begin{cases} x_2 = y_1 - x_1 \\ \boxed{y_2 = x_5 - x_1} \\ \boxed{x_6 = -x_1} \\ x_3 \; x_4 \; y_3 \quad \text{arbitrary} \end{cases}$$

(7)

$$\begin{cases} x_2 = y_1 - x_1 \\ y_2 = y_5 - x_1 \\ \boxed{y_6 = x_6 - x_1} \\ \boxed{x_7 = -x_1} \\ x_3 \; x_4 \; x_5 \; y_3 \; y_4 \; y_7 \end{cases}$$

(8)

$$\begin{cases} x_2 = y_1 - x_1 \\ y_2 = z_1 - x_1 \\ z_2 = y_6 - x_1 \\ y_7 = x_7 - x_1 \\ \boxed{x_8 = -x_1} \\ x_3 \; x_4 \; x_5 \; x_6 \\ y_3 \; y_4 \; y_5 \quad \text{arbitrary} \\ z_3 \; z_4 \\ y_8 \end{cases}$$

(9)

$$\begin{cases} x_2 = y_1 - x_1 \\ y_2 = z_1 - x_1 \\ z_2 = z_6 - x_1 \\ z_7 = y_7 - x_1 \\ y_8 = x_8 - x_1 \\ \boxed{x_9 = -x_1} \\ x_3 \; x_4 \; x_5 \; x_6 \; x_7 \\ y_3 \; y_4 \; y_5 \; y_6 \\ z_3 \; z_4 \; z_5 \\ z_8 \; z_9 \\ y_9 \end{cases}$$

if $n > 4$ $\quad x_n = -x_1$

at $n = 4$ $\quad x_2 = -x_1$

if $n > 6$ $\quad y_{n-1} = x_{n-1} - x_1$

at $n = 6$ $\quad y_2 = x_{n-1} - x_1$

if $n > 8$ $\quad z_{n-2} = y_{n-2} - x_1$

at $n = 8$ $\quad z_2 = y_{n-2} - x_1$

n	X	Y	Z	W	# groups
4	$\boxed{2}$	0	0		1
5	5	0	0		1
6	6	$\boxed{3}$	0		2
7	7	7	0		2
8	8	8	$\boxed{4}$		3
9	9	9	9		3
10	10	10	10	5	4
11	11	11	11	11	4

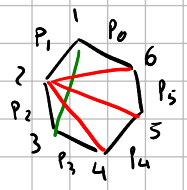
$$x_2 = -x_1 \rightarrow \begin{cases} x_2 = x_4 - x_1 \\ x_5 = -x_1 \end{cases} \qquad 4 \rightarrow 5$$

$$y_2 = x_5 - x_1 \rightarrow \begin{cases} y_2 = y_5 - x_1 \\ y_6 = x_6 - x_1 \end{cases} \qquad 6 \rightarrow 7$$

$$z_2 = y_6 - x_1 \rightarrow \begin{cases} z_2 = z_6 - x_1 \\ z_7 = y_7 - x_1 \end{cases} \qquad 8 \rightarrow 9$$

X_{ij} $\quad X_{rs}$ $\quad 1 \le i < j \le n$

$1 \le r < s \le n$



X_{13} crosses X_{24} X_{25} X_{26}

$$B = \sum_{i_1<j_1} \sum_{i_2<j_2} \cdots \sum_{i_N<j_N} \sum_{\substack{\alpha \in \mathbb{N}_0 \\ |\alpha| = N}} a_\alpha \prod_{k=1}^{N} \frac{1}{X_{i_k j_k}^{\alpha_k}}$$

$n=6$

$$\sum_{i \le j \le k} \frac{a_{ijk}}{x_i x_j x_k} + \sum_{\substack{i \le j \\ s}} \frac{b_{ijs}}{x_i x_j y_s} + \sum_{\substack{i \\ s,t}} \frac{c_{ist}}{x_i y_s y_t} + \sum_{r,s,t} \frac{d_{rst}}{y_r y_s y_t}$$

$$\sum_{\alpha,\beta} \frac{a_{\alpha\beta}}{x^\alpha y^\beta} = \sum_{\alpha_1 \alpha_2 \alpha_3 - \alpha_N} \frac{a_{\alpha_1 \alpha_2 - \alpha_N}}{X_1^{\alpha_1} X_2^{\alpha_2} - X_N^{\alpha_n}}$$